

Yan Ru Pei

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Research Profile

Research scientist and engineer specializing in efficient AI computation, distributed inference, temporal neural networks, and state-space models, with a foundation in statistical physics and optimization.

Professional and Research Appointments

Databricks, Inc.

Senior Software Engineer

Oct. 2026 – Present

Irvine, California (remote)

- Recruited through Databricks' specialized inference-engine hiring initiative to work on production large-language-model inference: request routing and scheduling, KV-cache memory and recovery, and deterministic profiling and simulation.

NVIDIA

Senior Deep Learning Algorithm Engineer

Feb. 2025 – Oct. 2026

Irvine, California (remote)

- Led design and implementation of Dynamo's KV-aware routing components, including replicated router state, standalone indexing, and runtime-independent selection services.
- Developed optimized KV-indexing implementations and comparative benchmarks for Dynamo, including concurrent radix-tree and positional approaches.
- Developed trace replay, GPU-free inference simulation, routing benchmarks, and configuration sweeps across Dynamo, DynoSim, and AISimulate, including AI Configurator integration.

BrainChip, Inc.

Senior Machine Learning Scientist (July 2024 – Feb. 2025)

Machine Learning Scientist (Oct. 2021 – July 2024)

Oct. 2021 – Feb. 2025

Laguna Hills, California

- Co-invented temporal event-based neural networks (TENNs), including buffered and recurrent architectures supported by BrainChip's [Akida 2](#) and [Akida Pico](#) processor IP.
- Developed PLEIADES temporal kernels, Centaurus tensor-contraction methods, and aTENNuate speech-enhancement models; first- or sole-authored the associated research papers.
- Developed causal event representations and a lightweight eye-tracking network for the bigBrains team's third-place result in the AIS 2024 challenge.

University of California, San Diego

Graduate Student Researcher, Physics

Sept. 2017 – Sept. 2021

San Diego, California

Researched memory-driven dynamics, optimization, and mode-assisted machine learning with Massimiliano Di Ventra.

Massachusetts Institute of Technology

Research Support Associate

Sept. 2016 – Aug. 2017

Cambridge, Massachusetts

Developed magnetic-field-control instrumentation for dysprosium magneto-optical trapping experiments in Wolfgang Ketterle's group.

University of California, Los Angeles

Undergraduate Research Assistant

Dec. 2015 – Sept. 2016

Los Angeles, California

Studied statistical unfolding in high-energy physics with Robert Cousins and simulated ion-beam collimation with Eric Hudson.

Education

- University of California, San Diego** 2017 – 2021
 Ph.D. in Physics, Computational Science Specialization. Dissertation: *Glassy Memory*.
- University of California, Los Angeles** 2014 – 2016
 B.S. in Physics and Applied Mathematics, *summa cum laude*.

Peer-Reviewed Publications

1. Tiyasa Mitra, Ritika Borkar, Nidhi Bhatia, Shivam Raj, Hongkuan Zhou, **Yan Ru Pei**, et al. [Beyond the Buzz: A Pragmatic Take on Inference Disaggregation](#). *Proceedings of Machine Learning and Systems*, vol. 8, pp. 2001–2015, 2026.
2. **Yan Ru Pei** and Olivier Coenen. [PLEIADES: Building Temporal Kernels with Orthogonal Polynomials](#). *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
3. **Yan Ru Pei**, Ritik Shrivastava, and Fnu Sidharth. [Optimized Real-time Speech Enhancement with Deep SSMS on Raw Audio](#). *Interspeech*, pp. 51–55, 2025. Introduces aTENNuate.
4. Qinyu Chen, Chang Gao, Min Liu, Daniele Perrone, **Yan Ru Pei**, et al. [Event-based Eye Tracking. Even-based Vision Workshop 2025](#). *IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pp. 5203–5215, 2025.
5. **Yan Ru Pei**. [Let SSMS be ConvNets: State-space Modeling with Optimal Tensor Contractions](#). *International Conference on Learning Representations (ICLR)*, 2025. Spotlight; sole author.
6. **Yan Ru Pei**, Sasskia Brüers, Sébastien Crouzet, Douglas McLelland, and Olivier Coenen. [A Lightweight Spatiotemporal Network for Online Eye Tracking with Event Camera](#). *IEEE/CVF CVPR Workshops*, pp. 5780–5788, 2024.
7. Zuowen Wang, Chang Gao, Zongwei Wu, Marcos V. Conde, Radu Timofte, Shih-Chii Liu, Qinyu Chen, et al., including **Yan Ru Pei**. [Event-Based Eye Tracking. AIS 2024 Challenge Survey](#). *IEEE/CVF CVPR Workshops*, pp. 5810–5825, 2024.
8. **Yan Ru Pei** and Massimiliano Di Ventra. [Non-equilibrium Criticality and Efficient Exploration of Glassy Landscapes with Memory Dynamics](#). *Physica A*, vol. 591, article 126727, 2022.
9. Sean R. B. Bearden*, **Yan Ru Pei***, and Massimiliano Di Ventra. [Efficient Solution of Boolean Satisfiability Problems with Digital Memcomputing](#). *Scientific Reports*, vol. 10, article 19741, 2020. *Equal contribution, designated in the published supplement.
10. Haik Manukian*, **Yan Ru Pei***, Sean Bearden, and Massimiliano Di Ventra. [Mode-assisted Unsupervised Learning of Restricted Boltzmann Machines](#). *Communications Physics*, vol. 3, article 105, 2020. *Equal contribution; Pei is also a corresponding author.
11. **Yan Ru Pei**, Haik Manukian, and Massimiliano Di Ventra. [Generating Weighted MAX-2-SAT Instances with Frustrated Loops: An RBM Case Study](#). *Journal of Machine Learning Research*, vol. 21, no. 159, 2020.
12. **Yan Ru Pei**, Fabio L. Traversa, and Massimiliano Di Ventra. [On the Universality of Memcomputing Machines](#). *IEEE Transactions on Neural Networks and Learning Systems*, vol. 30, no. 6, 2019.

Selected Preprints

- **Yan Ru Pei**. [At Most Two Infinite Blue Clusters in the CMR Representation of the Edwards-Anderson Spin Glass](#). arXiv:2605.17338, 2026.

- **Yan Ru Pei.** [peapods: A Rust-Accelerated Monte Carlo Package for Ising Spin Systems](#). arXiv:2602.19045, 2026.
- Yuqing Wang and **Yan Ru Pei.** [The Optimal Deterrence of Crime: A Focus on the Time Preference of DWI Offenders](#). arXiv:1909.06509, 2019.

Patents and Published Patent Applications

Granted Patent

- **Methods and System for Improved Processing of Sequential Data in a Neural Network.** Co-inventor. Australian patent [AU 2024203116 B2](#), 2025. Related U.S. published application: [US 2024/0378055 A1](#).

Published Applications

1. **Methods and Systems for Radar Signal Processing Using Temporal Neural Networks with State-Space Model.** Co-inventor. [US 2026/0266954 A1](#). Acknowledges U.S. Air Force support.
2. **Method and System for Implementing Temporal Convolution in Spatiotemporal Neural Networks.** Co-inventor. [WO 2023/250093 A1](#); [US 2025/0371319 A1](#); [AU 2023289134 A1](#); [EP 4544456 A1](#); [KR 20250040952 A](#).
3. **System of Interconnected State Space Models with Jointly Driven State Matrices.** Co-inventor. [WO 2026/148029 A1](#); [US 2026/0187189 A1](#).
4. **System and Method for Efficient Execution of Large Generative Artificial Intelligence Models on Edge Devices Using State-Space Models.** Co-inventor. [WO 2026/059996 A1](#); [US 2026/0072920 A1](#).
5. **Real-time Speech Enhancement on Raw Signals with Deep State-space Modeling.** Co-inventor. [US 2026/0179635 A1](#).
6. **Method and System for Implementing Encoder Projection in Neural Networks.** Co-inventor. [WO 2025/137663 A1](#); [US 2025/0209313 A1](#).

Selected Technical Publications

- Yongming Ding, **Rudy Pei**, Hongkuan Zhou, Ryan Olson, Alec Flowers, and Vikram Sharma Mailthody. [DynoSim: Simulating the Pareto Frontier](#). *NVIDIA Technical Blog*, May 29, 2026.
- **Rudy Pei**, John Thomson, Janelle Cai, Alec Flowers, Ryan Olson, and Dan Gil. [Flash Indexer: A Story of Inter-Galactic KV Routing](#). *NVIDIA Dynamo Documentation*, February 2026.
- Tony Lewis, Olivier Coenen, and **Yan Ru Pei.** [Introducing TENN: Revolutionizing Computing with an Energy Efficient Transformer Replacement](#). *BrainChip*; republished by the *Edge AI and Vision Alliance*, May 14, 2024.

Selected Open-Source Software

- **Dynamo and AISimulate.** Design and implementation of KV-aware routing, indexing, trace replay, and inference-simulation workflows. Public design issues and merged contributions are available under [PeaBrane](#).
- **mamba-tiny.** Developed readable PyTorch cumulative-sum and log-space implementations of the selective recurrence. The scan implementation is credited in the S2P2 code integrated into Ant Research's [EasyTemporalPointProcess](#) library.

- **peapods**. Developed a Rust simulation core with Python bindings for Ising and spin-glass systems, including Monte Carlo sampling, parallel tempering, replica-cluster moves, and reproducible experiment workflows.

Professional Recognition and Membership

- **KDD 2025 Outstanding Reviewer**, Applied Data Science Track, February cycle. Recognition awarded to the top 10% of reviewers in the cycle.
- **ICLR 2025 Spotlight**, sole-authored paper *Let SSMs be ConvNets*.
- **Third Place, AIS 2024 Event-Based Eye Tracking Challenge**, bigBrains team, BrainChip; associated paper published at CVPR Workshops 2024.
- **IEEE Senior Member**, elevated July 2024.

Speaking Invitations

Invitation to speak, [EdgeAI4R: Edge AI for Robotics, Emerging Technologies and Applications](#), IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), October 24, 2025.

Senior Program Committee and Area-Chair Service

- **Senior Area Chair**, KDD 2026 Applied Data Science Track, Cycle 1.
- **Area Chair**, ICASSP 2026: nine-paper portfolio.
- **Area Chair**, WCCI 2026, IJCNN Track: nine papers handled through final decision.
- **Area Chair**, IJCNN 2025: 20 meta-reviews.

Journal Refereeing

164 completed journal review rounds across 19 journals, through September 18, 2026. Counts include original and revised manuscript rounds; one item is a co-review.

Journal	Rounds
ACS Omega	1
Applied Intelligence	108
Digital Humanities and Society Studies	1
Environmental Research Letters	2
IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing	1
IEEE Transactions on Audio, Speech and Language Processing	1
IEEE Transactions on Emerging Topics in Computational Intelligence	5
IEEE Transactions on Neural Networks and Learning Systems	10
International Journal of Circuit Theory and Applications	3
International Journal of Image and Data Fusion	1
Journal of Environmental Engineering and Landscape Management	4
Land Degradation & Development	3
Machine Learning	1
Machine Learning: Science and Technology (co-review)	1
PLOS ONE	5
Remote Sensing Letters	4
Society & Natural Resources	2
The Journal of Physical Chemistry	2
Transactions on Machine Learning Research	9
Total	164

Applied Intelligence service comprises 108 completed rounds across 75 manuscripts.

Conference and Workshop Reviewing

161 completed conference/workshop paper reviews through September 18, 2026. Counts are per paper within each venue/cycle. Chair portfolios are listed separately on the preceding page.

Venue / cycle	Papers
AAAI 2026 Workshop: XAI4Science	5
ACM TheWebConf 2026 Industry Track	2
Asian Conference on Machine Learning 2026	3
CVPR 2025 Event-based Vision Workshop	2
ECAI 2025	5
Graph Signal Processing Workshop 2025	3
ICASSP 2025	4
ICASSP 2026	16
IEEE Spoken Language Technology Workshop 2026	4
WASPAA 2025	1
WCCI 2026, IJCNN Track	5
ICLR 2025 Workshops (four workshops)	9
ICLR 2026 Workshop: DeLTa	4
ICML 2026 Workshop: FoGen	5
IJCAI 2025	8
IJCAI-ECAI 2026	6
IJCNN 2025	8
ISCAS 2026	5
Interspeech 2025	7
Interspeech 2026	6
KDD 2025 ADS February cycle	3
KDD 2026 ADS Cycle 1	6
KDD 2026 ADS Cycle 2	5
KDD 2027 ADS Cycle 1	8
NeurIPS 2025 Datasets and Benchmarks	4
NeurIPS 2025 Workshop: CoML	3
NeurIPS 2026	6
NeurIPS 2026 Workshop: XAI4Science	3
PAKDD 2026	7
SysMus25	2
WSDM 2027	6
Total	161

Graduate Teaching

University of California, San Diego

- Physics 212C: Quantum Mechanics III, Spring 2019.
- Physics 200B: Theoretical Mechanics II, Winter 2019.
- Physics 243: Stochastic Methods, Fall 2018.

Accepted Internal Conference Contributions

NTECH US 2026, NVIDIA’s internal engineering conference, Santa Clara, October 26–29, 2026. Both contributions accepted September 16, 2026.

- **DynoSim – Simulating the Pareto Frontier of the Dynamo LLM Serving Stack.** Coauthor; accepted for an oral presentation. The program committee reports 160 oral acceptances from 900 submissions.
- **AI Sim: Lightning-Fast Offline LLM Serving Simulation.** Coauthor; accepted for a poster as *AIConfigurator Lightning-Fast Configuration Optimization for NVIDIA Dynamo LLM Serving*. The program committee reports 50 poster acceptances from 900 submissions.

Selected Media Coverage

- [BrainChip’s IP for Targeting AI Applications at the Edge.](#)
EE Times, hosted by Sunny Bains, February 7, 2025. Interview and transcript identifying Yan Ru (Rudy) Pei as a TENN inventor and discussing temporal modeling, speech enhancement, and event-based vision.
- Mike Falter. [BrainChip Accelerates AI at the Edge With New Low-Power Neural Processor.](#)
All About Circuits, October 29, 2024. Engineering coverage of Akida Pico and TENN-based low-power processing.
- Max Maxfield. [Mind-Boggling Neuromorphic Brain Chips \(Part 2\).](#)
EEJournal, March 9, 2023. Coverage of Akida 2.0 and TENNs for temporal vision, audio, and vital-sign applications.

Selected Technical Coverage and Implementation

- Abu Qader, Michael Feil, and Rachel Rapp. [2x Faster Inference with KV Cache-Aware Routing.](#)
Baseten Technical Blog, October 16, 2025. Baseten’s evaluation of Dynamo KV cache-aware routing reports 50% lower average time to first token and 34% lower time per output token in a high-load stress test, and 48% lower P95 and 49% lower P99 latency when shadowing OpenRouter production traffic. It also reports 61% more requests processed per second overall.
- Gcore AI Team. [Gcore Introduces Global Inference Routing Accelerated by NVIDIA Dynamo.](#)
Gcore, July 20, 2026. Reports extending Dynamo’s radix-tree KV matching in a 13-site experiment using real WAN links and GPU-free serving emulators, with 48% higher throughput under saturation. Gcore’s [July 23 public acknowledgment](#) names Rudy Pei among the NVIDIA collaborators on multi-datacenter KV routing and the DC Relay.
- Michael Dunn-O’Connor, Brian Zhang, and Shouzheng Liu. [Three Trends from MLSys 2026.](#)
Modular, May 29, 2026. Discusses the coauthored *Beyond the Buzz* paper and its analysis of inference disaggregation.
- [Akida Examples 2.15.0-doc-1.](#)
BrainChip software release, August 18, 2025. Introduced **PleiadesLayer** into the Akida toolchain and reported a 1.4-fold eye-tracking speedup and 54% hardware-memory reduction.

Technical Expertise

Research: machine learning training and inference; temporal and state-space models; neuromorphic computing; stochastic and dynamical systems; combinatorial optimization.

Systems: Rust, Python, distributed routing and caching, trace-driven simulation, GPU programming, quantization, and streaming inference.

Scientific software: PyTorch, JAX, TensorFlow, Triton, NumPy/Numba, and high-performance Monte Carlo simulation.